

Berkane Lab — Python Guided Tutorial

English practice book · 5 worked examples + 10 solved exercises per chapter

This companion manual is designed for independent practice. For each chapter, read the context, predict the output, inspect the execution trace, then compare your reasoning with the corrected Python solution. The examples and exercises are deliberately small: the objective is to make the data flow visible before increasing complexity.

Author: M. Haithem BERKANE · **Format:** static bilingual tutorial companion · **Palette:** Berkane Lab navy / burnt orange

Python foundations

Five worked examples

undefined

```
prenom = "Lina"  
print("Bienvenue", prenom)
```

Walkthrough: undefined

undefined

```
age = int("16")  
print(age + 2)
```

Walkthrough: undefined

undefined

```
a = float("12.5")
b = float("7.5")
print(a + b)
```

Walkthrough: undefined

undefined

```
valeur = 12
print(type(valeur).__name__)
```

Walkthrough: undefined

undefined

```
texte = "42"
nombre = int(texte)
print(nombre, nombre * 2)
```

Walkthrough: undefined

Ten solved exercises

undefined

Task: undefined

```
ville = input("Ville : ")
print("Bienvenue à", ville)
```

Correction: undefined

undefined

Task: undefined

```
age = int(input("Age : "))  
print(age + 1)
```

Correction: undefined

undefined

Task: undefined

```
c = 20  
print(c * 9 / 5 + 32)
```

Correction: undefined

undefined

Task: undefined

```
longueur, largeur = 5, 3  
print(longueur * largeur)
```

Correction: undefined

undefined

Task: undefined

```
print(17 % 5)
```

Correction: undefined

undefined

Task: undefined

```
print("Lina" + " · Oran")
```

Correction: undefined

undefined

Task: undefined

```
print(round(12.567, 2))
```

Correction: undefined

undefined

Task: undefined

```
print(12 > 0)
```

Correction: undefined

undefined

Task: undefined

```
a, b = 2, 5  
a, b = b, a
```

Correction: undefined

undefined

Task: undefined

```
score = 15  
print("Score :", score, "/ 20")
```

Correction: undefined

Conditions and loops

Five worked examples

undefined

```
n = 18  
if n % 2 == 0: print("pair")  
else: print("impair")
```

Walkthrough: undefined

undefined

```
score = 14  
if score ≥ 10: print("réussi")  
else: print("à revoir")
```

Walkthrough: undefined

undefined

```
t = 28
if t < 10: cat = "froid"
elif t < 25: cat = "doux"
else: cat = "chaud"
```

Walkthrough: undefined

undefined

```
compteur = 0
for i in range(5): compteur += 1
print(compteur)
```

Walkthrough: undefined

undefined

```
total = 0
for n in range(1, 7):
    if n % 2 == 0: total += n
print(total)
```

Walkthrough: undefined

Ten solved exercises

undefined

Task: undefined

```
n = -2
print(n > 0)
```

Correction: undefined

undefined

Task: undefined

```
note = 11
print("admis" if note ≥ 10 else "refusé")
```

Correction: undefined

undefined

Task: undefined

```
n = 12
print(n % 3 == 0)
```

Correction: undefined

undefined

Task: undefined

```
v = 65
if v < 50: r="lente"
elif v ≤ 90: r="normale"
else: r="rapide"
```

Correction: undefined

undefined

Task: undefined

```
total = 0
for n in range(1, 6): total += n
```

Correction: undefined

undefined

Task: undefined

```
for _ in range(3): print("go")
```

Correction: undefined

undefined

Task: undefined

```
i = 1
while i ≤ 3:
    print(i); i += 1
```

Correction: undefined

undefined

Task: undefined

```
compte = 0
for n in [8, 12, 15]:
    if n > 10: compte += 1
```


Correction: undefined

undefined

Task: undefined

```
maximum = 0
for n in [4, 9, 2]:
    if n > maximum: maximum = n
```

Correction: undefined

undefined

Task: undefined

```
for n in [1, 4, 7]:
    if n == 4: break
```

Correction: undefined

Functions and execution frames

Five worked examples

undefined

```
def saluer(nom):
    return "Bonjour " + nom
print(saluer("Lina"))
```

Walkthrough: undefined

undefined

```
def carre(n): return n * n
print(carre(6))
```

Walkthrough: undefined

undefined

```
def ttc(prix, taux=0.2): return prix * (1+taux)
print(ttc(100))
```

Walkthrough: undefined

undefined

```
def valide(note): return 0 ≤ note ≤ 20
print(valide(18))
```

Walkthrough: undefined

undefined

```
def c_to_f(c): return c * 9 / 5 + 32
def chaud(f): return f ≥ 77
print(chaud(c_to_f(25)))
```

Walkthrough: undefined

Ten solved exercises

undefined

Task: undefined

```
def est_pair(n): return n % 2 == 0
```

Correction: undefined

undefined

Task: undefined

```
def moyenne(a,b): return (a+b)/2
```

Correction: undefined

undefined

Task: undefined

```
def plus_grand(a,b): return a if a >= b else b
```

Correction: undefined

undefined

Task: undefined

```
def remise(prix, taux=10): return prix*(1-taux/100)
```

Correction: undefined

undefined

Task: undefined

```
def phrase(nom, ville): return f"{nom} habite à {ville}."
```

Correction: undefined

undefined

Task: undefined

```
def secondes(minutes): return minutes*60
```

Correction: undefined

undefined

Task: undefined

```
def categorie(age):  
    if age < 18: return "mineur"  
    return "majeur"
```

Correction: undefined

undefined

Task: undefined

```
def total(valeurs): return sum(valeurs)
```

Correction: undefined

undefined

Task: undefined

```
assert est_pair(4)
assert not est_pair(5)
```

Correction: undefined

undefined

Task: undefined

```
def carre(n):
    """Renvoie le carré de n."""
    return n*n
```

Correction: undefined

Vectors and lists

Five worked examples

undefined

```
ventes = [12, 8, 15]
total = sum(ventes)
print(total)
```

Walkthrough: undefined

undefined

```
temperatures = [18, 22, 19, 25]
maximum = temperatures[0]
for t in temperatures:
    if t > maximum: maximum = t
print(maximum)
```

Walkthrough: undefined

undefined

```
notes = [8, 12, 15, 9]
reussites = []
for note in notes:
    if note ≥ 10: reussites.append(note)
print(reussites)
```

Walkthrough: undefined

undefined

```
etapes = ["A", "B", "C"]
print(etapes[::-1])
```

Walkthrough: undefined

undefined

```
notes = []  
if notes:  
    print(sum(notes) / len(notes))  
else:  
    print("Aucune note")
```

Walkthrough: undefined

Ten solved exercises

undefined

Task: undefined

```
valeurs = [2, 5, 8]  
print(sum(n % 2 == 0 for n in valeurs))
```

Correction: undefined

undefined

Task: undefined

```
valeurs = [7, 3, 9]  
minimum = valeurs[0]  
for v in valeurs:  
    if v < minimum: minimum = v  
print(minimum)
```

Correction: undefined

undefined

Task: undefined

```
for i in range(len(valeurs)):
    print(i, valeurs[i])
```

Correction: undefined

undefined

Task: undefined

```
valeurs = [1, 4, 7]
modifiees = [v + 2 for v in valeurs]
```

Correction: undefined

undefined

Task: undefined

```
sortie = []
for v in [2, 2, 3, 2]:
    if v not in sortie: sortie.append(v)
```

Correction: undefined

undefined

Task: undefined

```
valeurs = ["A", "B", "C"]
valeurs = valeurs[1:] + valeurs[:1]
```


Correction: undefined

undefined

Task: undefined

```
total = 0
for v in [-2, 4, -1, 6]:
    if v > 0: total += v
```

Correction: undefined

undefined

Task: undefined

```
valeurs = [4, 9, 2]
for i, v in enumerate(valeurs):
    if v == 9: print(i); break
```

Correction: undefined

undefined

Task: undefined

```
temperatures = [23, 18, 25]
print(sorted(temperatures))
```

Correction: undefined

undefined

Task: undefined

```
valeurs = [4, 10, 7]
print(max(valeurs) - min(valeurs))
```

Correction: undefined

Dictionaries

Five worked examples

undefined

```
profil = {"nom": "Lina", "ville": "Oran", "age": 16}
print(profil["ville"])
```

Walkthrough: undefined

undefined

```
stock = {"clavier": 8, "souris": 12}
stock["clavier"] -= 1
```

Walkthrough: undefined

undefined

```
contact = {"nom": "Ali"}
print(contact.get("telephone", "Absent"))
```

Walkthrough: undefined

undefined

```
notes = {"maths": 15, "python": 17}
for matiere, note in notes.items():
    print(matiere, note)
```

Walkthrough: undefined

undefined

```
stock = {"livres": 4, "cahiers": 9}
print(sum(stock.values()))
```

Walkthrough: undefined

Ten solved exercises

undefined

Task: undefined

```
profil = {"nom": "Sara"}
profil["email"] = "sara@mail.test"
```

Correction: undefined

undefined

Task: undefined

```
donnees = {"nom": "Ali", "temp": True}
del donnees["temp"]
```

Correction: undefined

undefined

Task: undefined

```
votes = {}
for nom in ["A", "B", "A"]:
    votes[nom] = votes.get(nom, 0) + 1
```

Correction: undefined

undefined

Task: undefined

```
stock = {"A": 2, "B": 8}
for nom, quantite in stock.items():
    if quantite < 5: print(nom)
```

Correction: undefined

undefined

Task: undefined

```
source = {"nom": "Lina", "age": 16, "secret": 1}
cible = {k: source[k] for k in ["nom", "age"]}
```

Correction: undefined

undefined

Task: undefined

```
codes = {"Oran": 31, "Alger": 16}
inverse = {code: ville for ville, code in codes.items()}
```

Correction: undefined

undefined

Task: undefined

```
notes = {"A": 12, "B": 16}
print(sum(notes.values()) / len(notes))
```

Correction: undefined

undefined

Task: undefined

```
profil = {"nom": "Maya"}
print(profil.get("ville", "inconnue"))
```

Correction: undefined

undefined

Task: undefined

```
classes = {}
for nom, classe in [("A", "2A"), ("B", "2A")]:
    classes.setdefault(classe, []).append(nom)
```

Correction: undefined

undefined

Task: undefined

```
fiche = {"id": "S1", "nom": "Lina"}  
for cle, valeur in fiche.items(): print(cle, ":", valeur)
```

Correction: undefined

Records

Five worked examples

undefined

```
eleve = {"id": "S1", "nom": "Maya", "notes": [12, 15]}
```

Walkthrough: undefined

undefined

```
note = 18  
valide = 0 ≤ note ≤ 20  
print(valide)
```

Walkthrough: undefined

undefined

```
eleves = [{"nom": "A", "note": 12}, {"nom": "B", "note": 8}]  
for eleve in eleves: print(eleve["nom"])
```

Walkthrough: undefined

undefined

```
eleve = {"nom": "Lina", "notes": [14, 16]}  
eleve["moyenne"] = sum(eleve["notes"]) / len(eleve["notes"])
```

Walkthrough: undefined

undefined

```
eleves = [{"id": "S1"}, {"id": "S2"}]  
for e in eleves:  
    if e["id"] == "S2": print(e)
```

Walkthrough: undefined

Ten solved exercises

undefined

Task: undefined

```
livre = {"titre": "Python", "annee": 2026}
```

Correction: undefined

undefined

Task: undefined

```
fiche = {"nom":""}  
print(bool(fiche.get("nom")))
```

Correction: undefined

undefined

Task: undefined

```
for e in eleves: e["moyenne"] = sum(e["notes"]) / len(e["notes"])
```

Correction: undefined

undefined

Task: undefined

```
admis = [e for e in eleves if e["moyenne"] ≥ 10]
```

Correction: undefined

undefined

Task: undefined

```
ids = [e["id"] for e in eleves]  
print(len(ids) == len(set(ids)))
```

Correction: undefined

undefined

Task: undefined

```
capteur = {"nom": "T1", "unite": "°C", "mesure": 21.4}
```

Correction: undefined

undefined

Task: undefined

```
m = 21.4  
print(0 ≤ m ≤ 50)
```

Correction: undefined

undefined

Task: undefined

```
for e in eleves: print(e.get("nom", "inconnu"))
```

Correction: undefined

undefined

Task: undefined

```
trouve = next((e for e in eleves if e["id"] == "S2"), None)
```

Correction: undefined

undefined

Task: undefined

```
valides = [e for e in eleves if e.get("moyenne", 0) ≥ 10]
print(len(valides))
```

Correction: undefined

Stacks

Five worked examples

undefined

```
pile = ["titre", "image"]
print(pile.pop())
```

Walkthrough: undefined

undefined

```
historique = ["accueil", "cours", "quiz"]
page = historique.pop()
print(page, historique[-1])
```

Walkthrough: undefined

undefined

```
pile = []
for symbole in "(":
    if symbole == "(": pile.append(symbole)
print(len(pile))
```

Walkthrough: undefined

undefined

```
pile = []
if pile: print(pile.pop())
else: print("Rien à annuler")
```

Walkthrough: undefined

undefined

```
actions = ["A", "B", "C"]
for _ in range(2): print(actions.pop())
```

Walkthrough: undefined

Ten solved exercises

undefined

Task: undefined

```
pile = []  
for action in ["A", "B", "C"]: pile.append(action)
```

Correction: undefined

undefined

Task: undefined

```
if pile: resultat = pile.pop()
```

Correction: undefined

undefined

Task: undefined

```
sommet = pile[-1] if pile else None
```

Correction: undefined

undefined

Task: undefined

```
print(len(pile))
```

Correction: undefined

undefined

Task: undefined

```
while pile: pile.pop()
```

Correction: undefined

undefined

Task: undefined

```
pile = list("PY")  
print(pile.pop() + pile.pop())
```

Correction: undefined

undefined

Task: undefined

```
pile = []  
for x in [1,2]:  
    pile.append(x); print(pile)
```

Correction: undefined

undefined

Task: undefined

```
page = historique[-1] if historique else "aucune"
```

Correction: undefined

undefined

Task: undefined

```
if len(pile) < 3: pile.append("nouveau")
```

Correction: undefined

undefined

Task: undefined

```
def depiler(pile):  
    return pile.pop() if pile else None
```

Correction: undefined

Queues

Five worked examples

undefined

```
file = ["A.pdf", "B.pdf"]  
print(file.pop(0))
```

Walkthrough: undefined

undefined

```
file = ["Nora", "Ali", "Maya"]  
servie = file.pop(0)
```

Walkthrough: undefined

undefined

```
file = []  
if file: print(file.pop(0))  
else: print("Attente vide")
```

Walkthrough: undefined

undefined

```
file = ["A", "B"]  
compteur = 0  
while file:  
    file.pop(0); compteur += 1  
print(compteur)
```

Walkthrough: undefined

undefined

```
file = ["A", "B"]  
file.append("C")  
print(file)
```

Walkthrough: undefined

Ten solved exercises

undefined

Task: undefined

```
file = []  
file.append("T1"); file.append("T2")
```

Correction: undefined

undefined

Task: undefined

```
ticket = file.pop(0) if file else None
```

Correction: undefined

undefined

Task: undefined

```
while file:  
    print(file.pop(0))
```

Correction: undefined

undefined

Task: undefined

```
print(len(file))
```

Correction: undefined

undefined

Task: undefined


```
file.insert(0, "URGENT")
```

Correction: undefined

undefined

Task: undefined

```
servis = 0
while file:
    file.pop(0); servis += 1
```

Correction: undefined

undefined

Task: undefined

```
def servir(file):
    return file.pop(0) if file else None
```

Correction: undefined

undefined

Task: undefined

```
file.append(file.pop(0))
```

Correction: undefined

undefined

Task: undefined

```
temps = [4, 7, 5]
print(sum(temps) / len(temps))
```

Correction: undefined

undefined

Task: undefined

```
file1 = ["A","B"]; file2 = ["C","D"]
file = file1 + file2
```

Correction: undefined

Trees

Five worked examples

undefined

```
arbre = {"valeur":"Projet", "enfants":[]}
print(arbre["valeur"])
```

Walkthrough: undefined

undefined

```
noeud = {"valeur":"README", "enfants":[]}
print(noeud["enfants"] = [])
```

Walkthrough: undefined

undefined

```
def visiter(n):  
    print(n["valeur"])  
    for enfant in n["enfants"]: visiter(enfant)
```

Walkthrough: undefined

undefined

```
def hauteur(n):  
    if not n["enfants"]: return 1  
    return 1 + max(hauteur(e) for e in n["enfants"])
```

Walkthrough: undefined

undefined

```
def contient(n, cible):  
    return n["valeur"] == cible or any(contient(e, cible) for e in n["enfants"])
```

Walkthrough: undefined

Ten solved exercises

undefined

Task: undefined

```
arbre = {"valeur": "A", "enfants": [{"valeur": "B", "enfants": []}, {"valeur": "C", "enfants": [{"valeur": "D", "enfants": []}]}
```

Correction: undefined

undefined

Task: undefined

```
def afficher(n):  
    print(n["valeur"])  
    for e in n["enfants"]: afficher(e)
```

Correction: undefined

undefined

Task: undefined

```
def compter(n):  
    return 1 + sum(compter(e) for e in n["enfants"])
```

Correction: undefined

undefined

Task: undefined

```
def feuille(n): return len(n["enfants"]) == 0
```

Correction: undefined

undefined

Task: undefined

```
def h(n):  
    return 1 if not n["enfants"] else 1 + max(h(e) for e in n["enfants"])
```

Correction: undefined

undefined

Task: undefined

```
trouve = contient(arbre, "Cours")
```

Correction: undefined

undefined

Task: undefined

```
def afficher(n, niveau=0):  
    print("  " * niveau + n["valeur"])
```

Correction: undefined

undefined

Task: undefined

```
def feuilles(n):  
    if not n["enfants"]: return 1  
    return sum(feUILles(e) for e in n["enfants"] )
```

Correction: undefined

undefined

Task: undefined

```
def ok(n): return n["valeur"].startswith("C")
```

Correction: undefined

undefined

Task: undefined

```
def afficher(n):  
    if n is None: return  
    print(n["valeur"])
```

Correction: undefined

Introduction to statistics

Five worked examples

undefined

```
notes = [8, 12, 14, 15, 19]  
print(sum(notes) / len(notes))
```

Walkthrough: undefined

undefined

```
temps = [18, 25, 20, 22, 35]
tries = sorted(temps)
print(tries[len(tries)//2])
```

Walkthrough: undefined

undefined

```
mesures = [4, 8, 10]
print(max(mesures) - min(mesures))
```

Walkthrough: undefined

undefined

```
mesures = [12.5, 9.8, 15.2]
print(min(mesures), max(mesures))
```

Walkthrough: undefined

undefined

```
notes = [10, 10, 10, 20]
print(sum(notes)/len(notes))
```

Walkthrough: undefined

Ten solved exercises

undefined

Task: undefined

```
notes = [10,12,14]
print(sum(notes)/len(notes))
```

Correction: undefined

undefined

Task: undefined

```
valeurs = sorted([2, 8, 4, 6])
mediane = (valeurs[1] + valeurs[2]) / 2
```

Correction: undefined

undefined

Task: undefined

```
valeurs = [3, 11, 7]
print(max(valeurs)-min(valeurs))
```

Correction: undefined

undefined

Task: undefined


```
valeurs = [4, None, 6]
propres = [v for v in valeurs if v is not None]
```

Correction: undefined

undefined

Task: undefined

```
donnees = [2, 4, 6, 8]
print(len(donnees))
```

Correction: undefined

undefined

Task: undefined

```
m = sum(notes)/len(notes)
print([n for n in notes if n > m])
```

Correction: undefined

undefined

Task: undefined

```
a = [10, 12]; b = [14, 16]
print(sum(a)/len(a) < sum(b)/len(b))
```

Correction: undefined

undefined

Task: undefined

```
notes = [10, 10, 12]
frequences = {}
for n in notes: frequences[n] = frequences.get(n, 0) + 1
```

Correction: undefined

undefined

Task: undefined

```
v = sorted([4, 8, 10, 12, 16])
print(sum(v)/len(v), v[len(v)//2], max(v)-min(v))
```

Correction: undefined

undefined

Task: undefined

```
m = sum(notes)/len(notes)
print("niveau global satisfaisant" if m ≥ 10 else "à retravailler")
```

Correction: undefined

Suggested study routine

1. Read one example without running it. 2. Write the expected result. 3. Execute or simulate the code. 4. Explain the state change in one sentence. 5. Complete the ten exercises. 6. Return to the first exercise you could not explain and rewrite it without copying.

Important note

This PDF is a learning companion. The browser version provides bilingual navigation, local quizzes, code traces, examples and exercises without an API, login or database.